# DESIGN MILESTONES

Office type coffee machine - shared space - importance of hygiene and efficiency - Self cleansing mechanism

# Define

**User definition:** People who use shared spaces like a work environment, universities, school canteens, lounges, cafes etc.

**AI' role:** Defining pain points and the usage scenario.

**Usage Scenario:**

- The office employee goes to get coffee on their way to their meeting.
- The machine cleans its pipes and the serving area for the user.
- The employee selects their coffee of choice and waits while it is being prepared.
- Then gets their coffee and goes to their meeting feeling more freshened and energized

**Likely pain points accepted:**

- The waiting line could be long and people could be late to their appointments.
- Fear of contamination, hygiene anxiety.
- The machine can be loud and noisy and disturb others.
- Temperature safety. (if spilled, high temperature coffee can cause accidents)
- There might be too many options to choose from regarding the coffee and beverage choices this might cause choice overload.

# Ideate

**AI's role:** Clarifier and expander for research.

**AI Response Summary:**
Hygiene & "Zero-Touch" Interaction
Post-pandemic, people are hyper-aware of shared touchpoints.

- Gesture or Voice Control: Allow users to select their brew via a mobile app or simple hand gestures to avoid touching a communal screen.
- Automated Steam Wands: For milk-based drinks, a wand that automatically purges and cleans itself after every use is a massive hygiene win.
- Antimicrobial Materials: Use copper-infused plastics or specialized coatings on high-touch areas like the drip tray and buttons

**Accepted Ideations:**

- Self cleansing mechanism
- Office type coffee machine - shared space - importance of hygiene and efficiency
- The user sees the cleaning process by starting it and the puff of air that comes from the foaming tube after it is cleaned.
- The form of the product has changed throughout my process. The milk foamer compartment is removable and the water compartment is opened from the side to be filled. The coffee beans are placed on the top when the user opens the lid there is a place for the coffee beans.

# Expand

**AI Response Summary:**

- The Water Path: From Reservoir to Pressure
- The Coffee Path: Grinding and Compacting
- The Milk Path: The Venturi Effect
- Cleaning and Waste
- The Aeration & Milk Path Mechanics
- The Self-Cleansing Mechanics - The "Steam Pulse" (The Flash Clean)
- The Hot Water Flush
- Hygiene-First Design Features

AI's Role: Expanding design options and researching technical propeties.

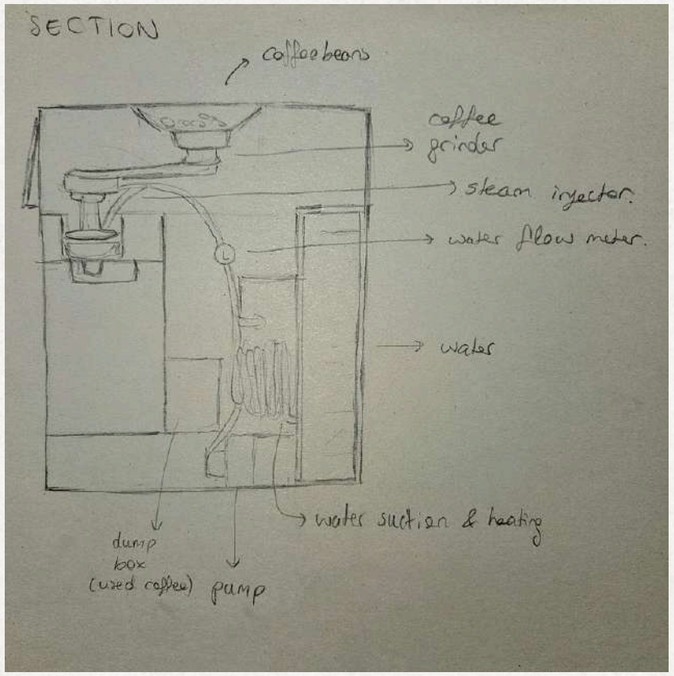
| Method | Mechanics | Frequency | Benefit |
|---|---|---|---|
| **Steam Purge** | High-pressure steam blast | After every drink | Prevents immediate clogs |
| **Hot Water Rinse** | Continuous flow of 90°C water | Every 2–4 hours | Dissolves fat deposits |
| **Chemical Cycle** | Detergent + Water circulation | Daily | Breaks down protein chains |

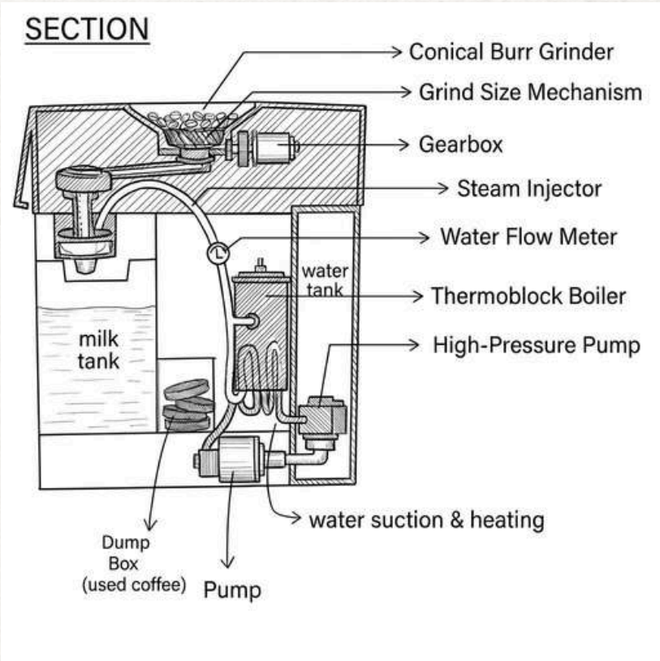
- **AI Design Tip:** One of the most effective ways to ensure a "clean feel" in a public environment is the Auto-Purge. By seeing a small puff of steam exit the machine after their drink is done, the user gets a visual confirmation that the machine is maintaining itself.
- **Reflection:** The information about self cleansing parts were taken into consideration and applied on the design.

**Key Insights:**

- The user sees the cleaning process by starting it and the puff of air that comes from the foaming tube after it is cleaned.
- The screen will display the need for cleansing as a notification after every use and the change of coffee and milk and water compartments when they are low will also be notified on the screen.

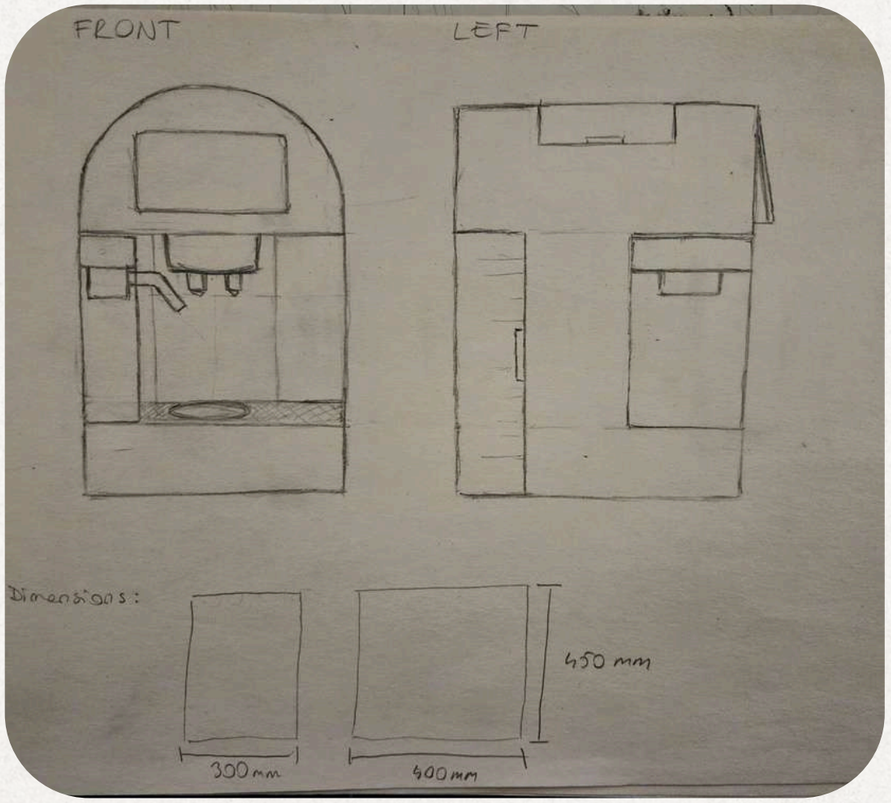
# Prototype




- Section drawing by hand was developed by AI to better show the technical properties of the coffee machine.

# Test

**AI's Role:** Generating 3D model and rendering.

- The chosen renders were developed using the modify features on vizcom.